Shiqi (Stella) Chen

Computer Science and Information Security Specialist

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Profile

Software enginner with experience in backend systems, internal platform engineering, infrastructure, and CI/CD automation in a production healthcare environment. Strong academic background in computer science and information security, with interests in low-level systems, cryptography and machine learning.

Education

University of Toronto Sep 2021 – Jun 2026

Honours Bachelor of Science with High Distinction Toronto, Canada

- Computer Science Specialist
- Information Security Specialist
- Mathematical Science Minor
- CGPA: 3.95/4.00
- Dean's List Scholar (2022 Summer, 2023 Winter, 2026 Winter)

The Affiliated High School of South China Normal University (HFI) Aug 2018 – Jun 2021

International Department, High School Guangzhou, China

- Focused on mathematics, physics, and computer science.
- President of the Rubik's Club.

Sun Yat-sen University Affiliated Middle School Sept 2015 – July 2018

Middle School Guangzhou, China

Professional Experience

PocketHealth May 2024 – Aug 2025

Software Engineering Intern (PEY) Toronto, Canada

- Developed Prometheus-based monitoring and internal alerting to track backend service health, traffic patterns, and operational reliability across distributed systems.
- Automated monthly Azure storage utilization reporting with cron-based pipelines, improving infrastructure cost visibility and reducing manual tracking for the finance team.
- Implemented JWT authentication to secure service-to-service communication across a distributed backend architecture.
- Designed and optimized dynamic GitLab CI/CD pipelines, modularizing infrastructure workflows and reducing deployment overhead.
- Maintained Kubernetes-based backend infrastructure on Azure, supporting containerized services, ingress controllers, deployment automation, and internal platform operations.
- Improved backend security and reliability by addressing vulnerabilities identified through automated security scanning and infrastructure review processes.
- Collaborated with platform and backend engineers to maintain deployment pipelines, troubleshoot

infrastructure issues, and improve reliability of containerized services in QA environment.

Kua.ai Jun 2023 – Aug 2023

Web Integration Intern Zhuhai, China

- Built an integration that connected Midjourney capabilities into the company's web platform despite the absence of an official public API.
- Applied reverse-engineering techniques to understand Discord-based API behavior and translate it into an internal integration workflow.
- Collaborated with teammates to refine implementation details and improve reliability of the final user-facing experience.

Academic Experience

University of Toronto Jan 2023 – Present

Teaching Assistant Toronto, Canada

- Served as a Teaching Assistant for undergraduate mathematics and computer science courses at the University of Toronto.
- Conducted tutorials, held weekly office hours, and provided instructional support to help students develop rigorous conceptual understanding and problem-solving ability.
- Supported course administration through invigilation and grading, with an emphasis on consistency, clarity, and fairness in assessment.
- Selected courses include MAT102, MAT135, MAT136, MAT223*, MAT224, MAT233, CSC236, CSC263, CSC347, and CSC427.

  * Served as Head Teaching Assistant (TA Coordinator).

University of Toronto Jan 2025 – Apr 2025

Research Assistant (Supervisor: Prof. Qun Wang) Toronto, Canada

- Built a model for the N-Vortex simulator to support simulation-based analysis and experimentation; validated and refined the implementation through testing, debugging, and iterative improvement.

University of Toronto Jan 2024 – Apr 2024

Lecture Notes Assistant (Supervisor: Prof. Lisa Jeffrey) Toronto, Canada

- Prepared and refined lecture notes in elementary Lie theory, and strengthened my background in Lie algebra through independent study of the underlying theorems and proofs.

Projects

Public-Key Cryptography and Digital Signature Analysis 2026

Cryptography, Number Theory, Public-Key Systems, Digital Signatures

- Analyzed algebraic and number-theoretic constructions underlying modern public-key cryptography, including Diffie-Hellman, ElGamal, RSA, DSA, and ECDSA.
- Studied the security assumptions behind discrete logarithm, integer factorization, and elliptic-curve based protocols, with attention to correctness, hardness assumptions, and implementation subtleties.

Street View Image Geolocation Predictor 2025

Computer Vision, Vision Transformers, Geolocation, Deep Learning

- Built a hierarchical multi-task geolocation model using CLIP ViT-B/32 to predict both city-level location and fine-grained local grid cells from four-directional Google Street View images.
- Designed a multi-view fusion architecture with multi-head self-attention to combine north, east, south, and west street-scene embeddings into a unified spatial representation.
- Compared raw and semantically masked image variants using DeepLabV3 preprocessing, showing that preserving transient visual context improved localization performance.
- Improved final model performance through deeper CLIP fine-tuning and loss balancing, achieving 79.0% city accuracy and 44.7% top-3 grid accuracy.

Chest X-ray Classification 2025

Machine Learning, Computer Vision, Semi-supervised Learning

- Built a semi-supervised deep learning pipeline using MoCo pretraining and Mean Teacher pseudo-labeling on NIH ChestX-ray14.
- Optimized training with torch.compile, TensorFloat-32, channels-last memory format, and local data caching to improve experiment throughput.

PBFT Smart Contract Verification 2025

Distributed Systems, Formal Methods, Blockchain

- Modeled PBFT consensus behavior using TLA+ to reason about safety and liveness properties.
- Analyzed protocol assumptions and failure cases in the context of smart contract infrastructure.

Virtualization Security and VM Escape Analysis 2024

Computer Security, Systems Security, Vulnerability Analysis

- Studied virtualization security through VM detection, anti-detection, and sandbox evasion techniques, focusing on how guest environments interact with hypervisor-level isolation.
- Analyzed QEMU VM escape vulnerabilities CVE-2015-5165 and CVE-2015-7504 using Debian as the guest environment, examining exploit conditions, attack surfaces, and containment failures.

Technical Skills

Programming	Python, Go, Kotlin, Java, C, C++, C#, TypeScript, JavaScript, SQL, Bash, R, Solidity, HTML/CSS, LaTeX
Tools	.NET, Angular, Node.js, PyTorch, Scikit-learn, NumPy, Pandas, Kubernetes, Docker, Azure, AKS, Terraform, GitLab CI/CD, Nginx, Prometheus, Helm, Linux, Git
Core Areas	Backend development, CI/CD Automation, Computer Vision, Deep Learning, Security, Cryptography, Distributed Systems, Cloud Infrastructure
Languages	English, Mandarin, Cantonese